Z1POWER · SUGARLAND BESS · STR MEDICAL

Multi-Discipline Design Review Board Consensus Finding Register

Combined EMS + MBMU control enclosure — Rev J deliverable set

PROJECT	z1Power Sugarland BESS — 375 kW / 1,221 kWh, ERCOT
SCOPE REVIEWED	One-page ARCH E master, 8-sheet A3 set, two native Altium SchDoc editions, Rev J BOM, six documents, verification scripts
METHOD	Six discipline leads (power, controls, fire/life safety, codes & AHJ, drafting/document control, vendor ICD) → independent adversarial verifier per discipline → chair consolidation with own spot-check
BOARD DATE	2026-08-09
STATUS	FOR REVIEW ONLY — NOT FOR CONSTRUCTION — PE REVIEW REQUIRED

2	30	112	148/148
CRITICAL	MAJOR	RAISED	NETLIST VERIFIED

What the automated verification does and does not prove. All five verification scripts pass: the drawn geometry on every format reproduces the 148-net design intent exactly, and the Altium binary audit is clean. That eliminates transcription error. It does not test the netlist against the BOM, does not test the annotation layer, does not test whether the netlist is buildable, and says nothing about device sizing, coordination, listing, marking or vendor conformance. Every finding in this register passes all five tests.

STATUS OF THIS BOARD. This review board is a set of AI review agents applying PE-level rigor. It is **not** a licensed professional engineer, and nothing in this document is a professional engineering opinion, a certification, or an approval. No finding here — including those marked CONFIRMED and independently re-derived from the source files — substitutes for review and sealing by a Texas-licensed Professional Engineer. The board does not certify; a named human signs.

BOARD OF REVIEW — CONSENSUS OUTPUT

Z1POWER SUGARLAND BESS — COMBINED EMS + MBMU CONTROL ENCLOSURE, REV J

Chair: review board (6 discipline leads + 1 independent verifier). Date of board: 2026-08-09.

The netlist is the strongest part of this package and the annotation, protection-device specification, fire-alarm architecture and label/schedule layer are the weakest; that gap is the whole story of this review.

---

1. CHAIR'S OWN SPOT-CHECK RECORD

I re-derived the highest-consequence surviving findings against the actual files before seating them. All six checks passed; none was demoted on the basis of my own check.

#	CHECK	METHOD	RESULT
1	Netlist integrity	`kicad-cli sch export netlist --format kicadxml` → `nettestG.py`	98 components, 150 KiCad nets, **148/148 matched, 0 failures**, 2 intentional spare-port no-connects. PASS is real.
2	No hardwired fire→ESS trip (R-01)	Python enumeration of all 148 nets	IO1 occupies pins **1–7 only** (DO0/pin 8 in zero nets). Regex over all 148 net names for `EPO|TRIP|STOP|SHUT` returns **[]**. No conductor from FACP1 to X3/X4/X5, Q1 or Q2. **CONFIRMED.**
3	FACP not on a dedicated branch (R-02)	Netlist + text extraction of all three governing PDFs	`CTRL_L1 = [CB2.3, F6.1, F8.1]`; `FACP_AC_L = [F8.2, FACP1.1]`; `SEC_N` carries FACP1 + PSU1 + PSU2 + PSU3. Zero hits for "FIRE ALARM CIRCUIT", "dedicated branch", "110.16", "arc flash", "409.22" across the ARCH E, the 8-sheet A3 set and the WireSpec. **CONFIRMED.**
4	TB11 USLKG (R-06)	`buildF.py:412` + rendered PDF + BOM	A3 sheet 8 renders `CT LANDING TB (UT/USLKG - 18 POS)`. ARCH E renders `CT LANDING TB (18P)` — **clean**. BOM item 50 says `UT 4`. General note 1 covers only SLC/SUP/LOOP/CAN-SIGNAL — **it does not cover CT terminals**. Exposure is confined to the A3 set, which is why I seat this MAJOR and not CRITICAL. **CONFIRMED as scoped.**
5	Fuse schedule (R-05)	Component-value dump from `outG/netlist.xml` + filesystem search	F1–F3 = `KLKR 1A`; **F4–F17 carry no ampere rating** (`KLKR CPT PRI`, `KLKR PSU1`, `KLKR 'BMS1'` …). No fuse schedule exists in `outG/`, `outF/`, `outA/` or `pkgJ/`. The referenced `TEST-REPORT.md` exists only in `outB/`, `outC/`, `outD/` — Rev B–D. **CONFIRMED.**
6	Branch breaker voltage rating (R-04)	BOM read	Items 20 and 23: `Branch breaker 20A 1P UL489 DIN, C-curve | ABB/Eaton | -` and the 6 A equivalent — **no part number, no voltage rating, no AIC** — while item 21 on the adjacent line does say `Q2 … 400V-rated`. **CONFIRMED.**

2. CONSOLIDATED FINDING REGISTER

Deduplicated across disciplines. Severity is the board's, re-ranked on consequence.

CRITICAL — 2

R-01 — No hardwired, fail-safe fire-alarm interlock to the PCS. 100% of shutdown authority is software over one Ethernet switch. *Disciplines: fire (FLS-01), code (CODE-01), controls (CTL-03).* Path from confirmed fire to a de-energized 375 kW PCS is: FACP1 dry contact → IO1 DI0 (Moxa ioLogik, not UL 864 listed) → SW1 (single switch, accepted deviation) → CPU1/CPU2 EMS application → Modbus TCP (no authentication) → Kehua. Every link is software or network, and IO1 sits on `24V_RMS` off PSU3, the one 24 V rail with **zero ride-through**, while CPU1/CPU2/SW1 ride UPS1. IO1's only modelled output (DO0) is unconnected. The Kehua Cabinet-375 has an EPO input (`pcs_spec.txt` line 137) the panel never reaches. All three fire inputs are **normally-open** (`FACP1.20/22/24`) sharing **one common conductor** (`FACP_COM`), so a single broken common is indistinguishable from "no fire" and is unsupervised. IFC 1207.5.5 / NFPA 855 (2023) 9.6 require automatic ESS shutdown on fire detection; NFPA 72 10.4 requires life-safety-dependent functions to be performed by the fire alarm system. **Mitigations the board credits:** CATL cabinets open their own contactors internally (`manual306.txt` 779–790), and SIR1/STB1 fire locally on FACP battery regardless. The uncovered gap is the inverter and any FACP-initiated hardwired trip. **Action:** normally-energized FACP relay → interposing relay → Kehua remote-stop; re-wire the three DI to Form-C N/C with supervision and split the common; move fire-path I/O to a backed rail. A two-terminal path already exists and is unused: insert an N/E contact in series with LOOP1/2/3 at TB20–22.

R-02 — Fire alarm control unit is not on a dedicated branch circuit, and none of the required circuit identification exists. *Disciplines: code (CODE-02), fire (FLS-03), power (verifier-added), controls (verifier-added).* `CTRL_L1 = [CB2.3, F6.1, F8.1]` — FACP1 shares the CB2 15 A 2-pole control main with PSU1, and through the other pole with PSU2 and PSU3. NEC 760.121(B)/760.41(B): *"the branch circuit supplying the fire alarm equipment shall supply no other loads"* — four other loads. NFPA 72 (2022) 10.6.5.1 same. Additionally absent everywhere in the set: red identification, "FIRE ALARM CIRCUIT" marking, lock-on device, and the permanent OCPD-location placard at the FACP. The only device between CB2 and the FACP is a fuse, which is not a disconnecting means. **This is black-letter and un-waivable; it fails plan review as drawn.** The FACP retains its 24 h battery, so it is not an immediate life-safety failure — that is why it is CRITICAL-on-code rather than CRITICAL-on-hazard. **Action:** dedicated 1P 15 A UL 489 device off `SEC_L1` feeding F8/FACP1 and nothing else, red handle, lock-on, placard. Note the compounding condition the board records: even then the fire alarm branch is derived from a 500 VA CPT downstream of Q1, so opening the panel main drops the FACP to battery.

MAJOR — 30

R-03 — Enclosure A/C and 100 W heater are purchased but exist on no net, with no source, no OCPD, no conductor and no disconnect; T2 has no per-leg load analysis. *power (P-01), code (CODE-06), fire (FLS-09 adjacent).* BOM items 52–53 have an **empty Ref Des**; neither appears among the 98 refdes or on any of 9 drawing sheets. The only 120 V source is a 500 VA CPT whose **each 120 V half-winding is rated 250 VA / 2.083 A**, and the design's own basis note ("S1 demand ~124 W run; 500 VA covers FACP alarm") is a total-VA statement with no per-leg analysis. Leg allocation is adverse: `CTRL_L1` = PSU1 + FACP1, `CTRL_L2` = PSU2 + PSU3, so the FACP alarm-plus-recharge case lands entirely on L1 and any ARC-100 AC draw above ~1.9 A puts that half-winding over rating. No NEC 440.32/440.22 sizing and no 440.14 disconnect for the A/C. The board does **not** seat the "225% overloaded / compressor start collapses the control bus" chain as an as-drawn defect — the A/C is on no circuit at all, so that is a hypothetical about a decision nobody has made. What is a defect: two purchased loads with no circuit, and a transformer sized without a per-leg census. **Action:** dedicated source and OCPD per the selected unit's MCA/MOCP; model both as refdes; publish the T2 VA census per leg; re-balance F8/F9 across legs or specify 1000 VA.

R-04 — CB3–CB8: six single-pole breakers applied line-to-neutral on 380Y/220 V with no voltage rating, no part number and no interrupting rating. *power (P-10), code (CODE-10), drafting (DQ-09).* NEC 240.85: the device must be able to clear a **380 V line-to-line** fault; the commonly stocked 240 V DIN device is 58% over. Q2 on the adjacent BOM line *is* voltage-specified (400 V), which proves the omission is inconsistent, not a house convention. The blank part number makes shop substitution likely, and the absent kAIC is a direct input the SCCR study cannot proceed without. **Action:** specify by full part number, UL 489, minimum

480Y/277 V (or 400Y/230 V), with marked IR; add "380Y/220 V — USE 480Y/277 V RATED DEVICES ONLY" adjacent to the bank on sheet 4.

R-05 — No fuse or breaker schedule exists; 14 of 17 fuses are unrated; F4/F5 undefined and NEC 450.3(B) undemonstrable; KLKR is the wrong fuse class. *power (P-02), code (CODE-11), drafting (DQ-07), fire (verifier-added).* The drawings point to a fuse schedule; the BOM points back at the drawings; **no schedule exists** — a circular reference, and the referenced TEST-REPORT is Rev B–D vintage and not in the package. T2 primary FLA = 1.042 A, secondary rated = 2.083 A; CB2 at 15 A is **720%** of secondary, so the 125% gate in Table 450.3(B) is not met and primary-only applies: **F4/F5 ≤ 3 A**. Littelfuse KLKR is **fast-acting** Class CC and will not hold a 500 VA CPT's 12–25× magnetizing inrush; the same objection applies to F6/F7/F9 on switch-mode inputs. F13–F15 are the sharpest case: under FSS option A they must carry MBMU quiescent **plus 1.1375 A** of alarm current per cabinet, and sheet 3 says they carry over Carson labels sized for BMS logic only. **Action:** publish a fuse and breaker schedule as a controlled sheet (refdes, class, amps, voltage, IR, code basis); set F4/F5 at 2–3 A **time-delay** Class CC (CCMR/FNQ-R).

R-06 — TB11 still specified "UT/USLKG" on Rev J sheet 8; the Rev J terminal fix missed the 18-pole CT block. *power (P-03).* If a shop installs USLKG 5 there, all 18 poles of the nine 333 mV differential channels bond to the DIN rail and to PE — every channel shorted, meter reads zero on all nine. Drawing contradicts BOM (`UT 4`) and contradicts the Engineer Handbook ("green/yellow anywhere but SB1-3 = stop the build"). Confirmed by the chair; confined to the A3 set (ARCH E is clean), which caps it at MAJOR. **Action:** correct `buildF.py:412`; grep every value string for USLKG and confirm only SB1/SB2/SB3 survive; add a terminal-block schedule.

R-07 — Stale Rev C / Rev I content survives on the governing sheets, including an instruction to create a second N-G bond. *drafting (DQ-01, DQ-06), power (P-12), vendor (VI-13), code (verifier-added).* A3 sheet 2 still says "PAD-MOUNT AUX TX … FED FROM Q1 LOAD SIDE VIA TB12. 40A MAIN = NEC 450.3(B) PRIMARY PROT" for a transformer and terminal block deleted in Rev I (TB12 is not among the 98 refdes). ARCH E still says "EXTERNAL PAD TX: N-G BOND + GEC AT TX". **A3 sheet 1 says "N-G bonds on BOTH transformers"** — an explicit instruction to create the second bond NEC 250.30(A)(1) exists to prevent. Sheet 1's banner reads **REV F** under a Rev J title block, its note stack ends at "REV D ADDS", and open item (5) is a fabrication gate against deleted equipment. Consequence beyond the confusion: the obsolete note is now **the only 450.3(B) statement in a Rev J deliverable**, and it certifies the code question for a transformer that does not exist while T2 — which needs it — has none. This is a gate-6 change-control failure. **Action:** rebuild sheet 1 with a real revision-history table; delete the pad-TX strings from `buildF.py` and `buildG.py`; add a build-time check that every refdes token in note text exists in `golden_nets.json`.

R-08 — 24V_RMS is a single point of failure carrying every safety-relevant load, while the redundancy protects 26 W of EMS. *controls (CTL-02), power (P-08), fire (FLS-02), code (verifier-added).* `24V_RMS = [F13–F17, PSU3.4]` — one NDR-240-24, no ORing module, no UPS — feeds MBMU1-3 (all BMS telemetry), EM1-3 (all CATL Ethernet) and IO1 (the FACP interface). `24V_UPS = [F10, F11, F12, UPS1.3]` feeds only CPU1/CPU2/SW1 (26.4 W, 35% loaded, ~100 s hold-up) behind PSU1+PSU2+RM1+UPS1 (~$645). Loss of PSU3, F9, one CB2 pole, or leg L2 removes BMS visibility and the FACP monitoring input while the EMS stays up. UDGOK gates 3 and 4. **Note the reviewer's fix does not work as written:** PSU1/PSU2 are 3.2 A each and UPS1 is 5 A, both below the 3.41 A FSS alarm load alone. **Action:** PSU4 + a second ORing module dedicated to 24V_RMS, or diode-OR PSU3 into an upsized backed bus; move F16/F17 at minimum; add an explicit "panel AC lost" point on a backed input.

R-09 — No supply-health signal from PSU1/PSU2/PSU3/RM1/UPS1 reaches the EMS; the designed N+1 is an undetected N. *controls (CTL-10).* PSU1-3 appear only on pins 1–5, RM1 on 1–3, UPS1 on 1–4. No DC-OK, redundancy-fault, buffer-mode or end-of-hold-up contact exists in 148 nets. First failure is silent; the panel then runs single-thread until the second. Same for UPS1 — the ~100 s of hold-up expires with no warning and no orderly shutdown. Fix is three wires and spare I/O the design already owns. **Action:** wire DC-OK / redundancy-fault / UPS-alarm to spare DI; correct the IOLGK symbol to the E1214's real 6 DI / 6 DO so the spares are visible.

R-10 — Both mutually exclusive FSS architectures are live in the verified netlist; the machine-checked source of truth certifies a configuration that must not be built. *fire (verifier-added).* Option A (MBMU LOOP → TB20-22 → X20-22 → JX1.K/L) and Option B (FACP REL1-3 → TB14 → X12-14 pins 5/6) are both fully drawn and both counted in the 148 nets that all five scripts certify. The only thing preventing a shop from building both is a text note. **Action:** carry one architecture as live nets and the alternate as an explicitly flagged option, so the verification result means what it says.

R-11 — Class A SLC is home-run through the panel; outgoing and return legs of each cabinet share one 6-pin field interface. *fire (FLS-05).* Every hop is spliced at TB8/TB9/TB10, so all 18 fire conductors home-run and the panel→cab1 and cab1→panel legs sit in one raceway. One crushed conduit isolates that cabinet in both directions — the single fault Class A exists to survive. NFPA 72 (2022) 12.3.6. CATL's own JX1_IN/JX2_OUT C/D/E/F pass-through pins exist precisely so the fire pairs daisy-chain rack to rack; sheet 7's own note ("INTER-CABINET RUNS ARE FIELD WIRING") describes a topology the netlist contradicts. The board records that the CA-6075 still buys survivability for the two undamaged cabinets — it is not worthless — but the pathway is not Class A compliant. `Customer-Overview.pdf` tells the customer "ring-wired detectors survive a cut cable"; either the topology changes or the sentence does.

R-12 — The three cabinet 24 V FSS circuits are drawn as releasing outputs onto a pin function that does not exist, at 114% of the ARC-100 circuit rating, and the required aux supply is a BOM contingency only. *fire (FLS-04, FLS-09).* `SUP1/2/3` are FACP1 REL1-3 → TB14 → X12-14 pins 5/6. CATL's rack has **both** a continuous `Loop_Input_24V±` (JX1_IN 1/2) and a discrete `Aerosol Input±` (JX1_IN 3/4) — the design has collapsed them onto one pair and connects to neither aerosol pin. In standby the CATL sensor circuits would see 0 V. Load is **1.1375 A per cabinet = 114% of the ARC-100's stated 1 A per-circuit rating**, 3.4125 A total, 4.27 A at 125%. PSN-1000 is in the BOM at "Basis: TBD" and in **zero of the 148 nets** — so its dedicated 760.121(B) branch, its own NFPA 72 battery calculation, its AC-fail/trouble contact and its DIN/weight footprint are all missing. **Action:** resolve circuit type at the ICD but fix the drawing now; if the PSN is required, model it fully.

R-13 — Three purchased supervised-suppression EOL devices exist on no drawing and in no net. *fire (FLS-07).* BOM item 46 buys **4** Potter EOLD 3005012; the netlist has **1** (R4, on the NAC). SUP1/2/3 terminate at the cabinet field boxes with nothing at the far end, so an open in any cabinet's 24 V suppression run is undetectable — while the Engineer Handbook asserts "EOL devices supervise every circuit." Same class of defect as the Rev J USLKG finding: a life-safety supervision element that exists only in the BOM. **Action:** assign R5/R6/R7, place them, netlist them; EOL type follows from R-12.

R-14 — Abort: one unsupervised two-wire pair, one station, three hazards, no abort-active indication, no part number. *fire (FLS-08).* `ABORT_P/N` are two-node nets with no EOL and no supervision. A chafed or wetted short holds abort asserted and inhibits release in **all three** cabinets, silently. PB1 is on the control enclosure, not in any protected volume; NFPA 72 23.8.5.6 requires location within the protected area, constant manual pressure, and distinctive indication while in effect. BOM item 45 has a blank part number and does not establish the momentary constant-pressure characteristic. Additional unresolved point: under FSS option A the aerosol fires internally on dual-detector, so PB1's function is undefined until the option gate closes.

R-15 — No manual release station and no releasing-circuit maintenance disconnect. *fire (verifier-added).* Searching all 148 nets and all 98 components for any manual actuation or release-disable device returns nothing. Technicians will work inside three cabinets each armed with an aerosol generator, with no drawn means to disable release.

R-16 — NFPA 72 secondary-supply calculation is internally inconsistent across three documents, and CATL's 2 h notification term is not captured. *fire (FLS-10), drafting (DQ-13), vendor (VI-18).* Sheet 7 says **~3.3 Ah**; the BOM and the Handbook say **4.8 Ah**. The board re-derived the calculation from the Rev D inputs: 24 h × 0.15045 A = 3.611 Ah + 5 min × 3.9625 A = 0.330 Ah, ×1.2 = **4.73 Ah** — the drawing figure is 31% low and is wrong. Separately, `manual306.txt:809` requires "24h power supply for fire alarm **and 2h power supply for sound and light**"; only the 24 h half is on the drawing. At a 2 h alarm term the requirement is ~13.8 Ah against 8.0 Ah installed. **Action:** one signed worksheet referenced by number from all three documents; resolve the 2 h vs 5 min question with the AHJ (NEC 110.3(B) argues CATL governs).

R-17 — No NEC 409.110 nameplate, no 409.22(B) field marking, no 110.16 arc-flash label, no label schedule. *code (CODE-03, CODE-08, CODE-14).* Confirmed by the chair: zero hits for "arc flash", "110.16", "409.22", "enclosure type", "TYPE 4" or "60 Hz" across all six PDFs. Six of the seven 409.110 items are absent — responsible organization, per-supply voltage/phase/frequency/FLA (480 V FLA ≈ 1.3 A, aux ≈ 17.9 A/phase, neither stated), SCCR, wiring-diagram identification number, and enclosure type number. ESS signage (NFPA 855 / IFC 1207.5.4) likewise absent. **Action:** one label-schedule sheet — label ID, verbatim text, material, location, code citation.

R-18 — Aux feeder protection method is unstated, and the set contradicts itself on it. *power (P-05b), code (CODE-09).* Kehua's 30 kVA / 380 V secondary FLA = 45.6 A, so its three output breakers are plausibly 50–63 A. 10 AWG Cu is capped at **30 A** by NEC 240.4(D)(7) (32.8 A after the 0.82 ambient factor). Sheet 4 says only "CONDUCTOR PROTECTION PER KEHUA OUTPUT BREAKER RATING + 240.21" — no subsection, no length limit, no ampacity floor. Worse, the ARCH E says "SIZE AUX FEEDER TO KEHUA OUTPUT BKR RATING (AWG10 CU MIN)" — full-size — while the WireSpec says "10 AWG feeder" — tap. Two different design methods in one revision. **Action:** state the governing rule explicitly ("240.21(B)(2) 25 ft tap, max 25 ft, ampacity ≥ ⅓ of the Kehua breaker, terminating in Q2 as the single OCPD") plus a hard note that any Kehua breaker above 30 A requires an upsized feeder.

R-19 — Chiller branches sized by Article 210/430 instead of Article 440, and the printed justification names a device that is not installed. *power (P-06), drafting (DQ-02), code (CODE-16), vendor (VI-12).* CATL publishes mutually impossible numbers (3 kW **and** 10 A at 220 V implies PF 1.36), so no verified RLA exists. The design took 13.64 A, applied a generic 125%, and rounded to 20 A. For a hermetic motor-compressor, NEC 440.4(B) requires the nameplate MCA/MOCP and 440.22(A) caps the branch OCPD at 175% of rated-load current — 175% × 10 A = 17.5 A, and 440.22(A) has **no next-size-up allowance**, so 15 A would be the ceiling on CATL's own stated figure. NEC 440.14 (disconnect within sight) is unaddressed. Separately and independently confirmed by the chair: **both governing drawings still print "15A C-CURVE HOLDS EITHER (MAG PICKUP ≥75A)"** — the 5× figure for a 15 A device — a few lines from "CB3/5/7 UPSIZED 15A > 20A C-CURVE" and from `CB3 = 20A`. That stale sentence is the only trip-curve arithmetic a reader has. The board's engineering position: **the 20 A device is correct** (31 A worst-case surge = 1.55× In, far below the 100–200 A

C-curve band), and the defect is (a) the wrong code basis with no MOCP, and (b) a specification that names two different devices for one circuit.

R-20 — WAN lands on the control switch; the published VLAN note is wrong on 6 of 13 used ports and cites a port that does not exist. *controls (CTL-09), drafting (DQ-03).* Actual map: P1=WAN, P2/P3=CPU WAN, P4/P5=CPU LAN, P6=meter, P7=IO, P8-P10=PCS, P11-P13=CATL ETH, P14 spare. Sheet 5 says "VLAN 10 WAN (P1,P4,P5) / VLAN 20 LAN (P6-P12) / VLAN 30 BMS (P13-P15)" — it puts **both EMS LAN ports in the WAN VLAN**, leaves both EMS WAN ports in no VLAN, puts two CATL ETH modules in the LAN VLAN, and references **P15 on a 14-port switch**. Both interfaces of both UC-2222A routers return to the same fabric, so the routers' trust boundary is bypassed at Layer 2 and VLAN config is the sole barrier — configured from a note that is wrong. Kehua's Modbus TCP is unauthenticated, port 502, read-write registers. **Action:** generate the VLAN/port/IP table from `golden_nets.json`, make it a signed sheet with a stored switch config as a fabrication deliverable; take the WAN off SW1 or firewall it.

R-21 — No I/O point list, alarm schedule, or Modbus register/setpoint schedule; PCS comms-loss behaviour undefined. *controls (verifier-added, CTL-16).* Nothing in the set states, per point: refdes, signal, normal state, alarm state, delay, scaling, EMS tag. Consequence: R-01's DI polarity table has nowhere to live, R-08's assumption that the EMS alarms on loss of the I/O module is unverifiable, R-20's VLAN/IP assignments have no schedule, and no watchdog timeout or failsafe setpoint exists for a grid-interconnected resource whose only control path is Modbus TCP over one switch. All three Kehua ports also ship at the same default 192.168.28.240. For a controls enclosure the point list is a primary engineering deliverable.

R-22 — No 24 VDC load schedule; PSU3 was sized against an unenumerated base load. *controls (CTL-08).* The Rev J justification reads "3 × 1.1375 A = 3.41 A + base exceeded NDR-75 3.2 A" and the base is never enumerated. CATL publishes **no 24 VDC figure** for the MBMU or ETH module anywhere. Sensitivity: PSU3 runs anywhere from **56% to 96%** of its 10 A depending on six unknown devices — and the NDR-240-24 derates above 50 °C, which a Gulf Coast enclosure will see. This blocks R-16's battery sizing and R-23's coordination study. Gate-7 violation on the single number the Rev J fix was built on.

R-23 — 24 VDC branch fuses cannot be shown to clear; no coordination study exists. *controls (CTL-05).* The NDR family current-limits at 105–135% of rating, so PSU3 delivers ~10.5–13.5 A into a bolted short; UPS1 caps Rail A at 5 A. Class CC fuses need roughly 5–10× rating for fast clearing. A branch fault therefore collapses the shared rail instead of isolating — and 24V_RMS is already single-source (R-08). Ampacities are undocumented (R-05), so nobody can check it today. **Action:** publish the 24 VDC schedule and demonstrate I_available/I_fuse against the real NDR and QUINT current-limit characteristics and the real KLKR TCC; electronic circuit protectors are one valid answer, not the only one.

R-24 — FSS loop conductor specification is self-contradictory and unbuildable as written. *controls (CTL-07), vendor (VI-06, verifier-added).* The wire spec calls AWG16 for the FSS loop; two lines later it says "20AWG ALL CONNECTOR CRIMPS", and the ARCH E says both about **the same net** (`LOOP1_P`, MBMU J3.6 crimp → TB20 → JX1_IN crimp). AWG16 = 1.31 mm² against a 0.5–0.75 mm² contact range. BOM item 55 funds both. The engineering reason for AWG16 is real and quantified: at 1.1375 A over a 30 m one-way run, AWG20 drops 2.27 V (9.4%, arriving 21.7 V at the initiator) versus 0.90 V on AWG16. **Action:** pick one, show the transition at TB20-22 on the drawing, state the actual per-cabinet run length, and obtain CATL's minimum aerosol initiator actuation voltage.

R-25 — No NEC 760.136 separation requirement and no enclosure layout drawing. *fire (FLS-13), code (CODE-15), power (P-11).* One enclosure carries 480 V, 380Y/220 V, 120/240 V control and all SLC, NAC, releasing, abort and FSS conductors. Searching the entire 9-page wire spec and all 8 sheets returns **zero** hits for "separat", "barrier", "wireway", "raceway" or a 760 citation. NEC 760.136(D) permits the mix only with a fixed barrier or 0.25 in maintained separation; the only wiring-management BOM line is "DIN rail, wire duct, panel hardware." Because there is **no layout or elevation sheet anywhere in the set**, neither the separation, nor NEC 110.26 working space, nor 312.6 bending space at TB1 (8 AWG in UT 6), nor the "SPD leads ≤0.5 m" note can be reviewed at all.

R-26 — Fire field conductors: FPL is not listed for the wet locations these runs pass through, and no maximum run length is stated. *fire (FLS-14).* NEC 300.5(B) makes underground raceway a wet location and 310.10(C) requires wet-location-listed conductors; FPL is general-purpose. Length is unbounded: at 1.1375 A with a 10% budget the limits are 165 ft on AWG18, 263 ft on AWG16, 418 ft on AWG14, and the drawing permits 18 AWG with no stated maximum.

R-27 — No automatic detection at the fire alarm control unit location. *fire (FLS-11).* Zero detectors in the netlist and in the BOM, in an unoccupied outdoor enclosure containing 480 V switching, a 500 VA transformer, three SMPS, a supercapacitor bank and two SLA batteries. NFPA 72 10.4.4 (10.4.4.1 permits heat detection where ambient prohibits smoke — relevant at 49 °C). No disposition anywhere, not even a deferral note. Application to an equipment enclosure rather than a room is an AHJ interpretation.

R-28 — No listed means of off-premises transmission from an unattended ESS. *fire (FLS-12), code (CODE-13).* No DACT, no UL 864/UL 2572 IP or cellular communicator in 98 refdes or 57 BOM lines. The only egress is FACP contact → IO1 → EMS network, which is not listed fire alarm transmission equipment. IFC 1207.5.5 / NFPA 855 require the detection system to be monitored; NFPA 72 Ch. 26 governs the transmitter listing. A trouble condition on a releasing system can persist indefinitely with nobody notified.

R-29 — Zero surge protection on any field-entering signal circuit. *controls (verifier-added).* SPD1 on the AC feeder is the only SPD in 98 refdes and 57 BOM lines, while **17 field boxes** bring in six circuit families from outdoors: WAN, PCS Ethernet, CAN (20–30 m to separately grounded cabinets), Class A SLC and releasing circuits, 333 mV CT leads originating inside 480 V gear, and FSS 24 V release loops. Two separately grounded outdoor enclosures on the Gulf Coast with a shielded pair between them is the textbook GPR/lightning case (IEEE C62.41.2, NFPA 780 Annex). The releasing circuits are the worst exposure — an induced surge sits directly on aerosol initiator wiring.

R-30 — Open-neutral on the 120/240 V CPT secondary destroys the fire panel, and the shared neutral defeats the PSU redundancy. *power (verifier-added), controls (verifier-added).* `SEC_N = [BD1.1, FACP1.2, PSU1.2, PSU2.2, PSU3.2, T2.5]` — one neutral for all three supplies and the FACP AC input. Loss of that neutral opens no fuse and places the lightly loaded L1 leg (PSU1 + ARC-100) in series with the heavily loaded L2 leg across 240 V; the NDR family rides 85–264 VAC, the Potter ARC-100 does not. Most likely outcome of a loose lug at T2.5 or BD1: a destroyed fire alarm panel with all three PSUs still running. This is also a common-mode single point that defeats the deliberate L1/L2 PSU split. Related and cheap: PSU2 and PSU3 are on the same leg, so one leg loss reproduces the R-08 failure signature by a second independent mechanism.

R-31 — BOM: 30 of 57 lines have no orderable part number, UL 489 is asserted for two supplementary-protector families, and no interrupting rating appears anywhere. *drafting (DQ-09), code (CODE-04), power (P-13).* Q1 is described "UL489" and named "Phoenix Contact TMC S3D [VERIFY]"; CB1 is described "UL489" and named "ABB S203-K30" — an S200-family device. If CB1 is a supplementary protector, NEC 240.10 bars it from substituting for branch protection, and the 10 AWG SPD leads are then protected only by

Q1 at **40 A** against a 35 A (75 °C) ampacity — a plain NEC 240.4 violation independent of any listing argument. Q1's status matters more: it is the declared panel main OCPD under NEC 409.21. **Honesty caveat for the PE:** the UL 1077 attribution rests on catalogue knowledge, not on any datasheet in the project record — obtain the listing letters.

R-32 — LTE cellular backup is promised in four issued documents and exists in zero components and zero BOM lines. *controls (CTL-14), code (CODE-13), drafting (DQ-04).* The Engineer Handbook gives LTE a full equipment card and lists it in the panel block diagram; the Customer Overview and How-It-Works both state "cellular backup takes over in seconds." This is not a deferred line item — it is a capability affirmatively promised to the customer that the design cannot deliver. The simulator is the one artifact that discloses the gap, flagging it as an IFC add at ~$400–800. SW1 P14 is free, so the fix is cheap. **Build it or strike the claim in the same revision.**

R-33 — MCAN rack-end terminator and its JX2_OUT connector are neither procured nor coordinated, and no ≤30 m length budget exists. *vendor (VI-04), controls (CTL-01).* **The board records a genuine disagreement between two verifiers and resolves it as follows.** CATL's own continuity checklist requires the J902 9-12 / 10-11 wire jumpers to read "on" and requires **~60 Ω at J902.21/22** — two 120 Ω in parallel — which proves the bus is terminated at both ends with no customer-added resistor at J902. **Do NOT add 120 Ω resistors at J902; that would break CATL's acceptance test.** What survives is real: the External Wiring diagram draws the far-end 120 Ω as a discrete external assembly on the last rack's JX2_OUT, `manual306.txt` 5.7.2.1 requires "JX2_OUT is connected to the terminal resistor connector," each of the three cabinets is simultaneously first and terminal rack, and there is **no BOM line, no part number and no coordination note** for that assembly — while CATL's scope statement (`ext_wiring.txt:135`) assigns external cables and any CANbridge to the customer. Nor is there any length calculation against the 30 m MBMU-to-terminator limit the drawing itself quotes.

R-34 — Neither incoming supply has an externally operable or lockable disconnect provision, and there is no emergency shutoff. *code (verifier-added).* Q1 and Q2 are DIN-rail devices on the backplate. No through-door handle, flange disconnect, lock-off device or door interlock in 57 BOM lines or any drawing — and no layout sheet on which one could be shown. NEC 110.25 requires lockable-open capability where lock-out is required; R-02's fire circuit needs a lock-**on** provision on the same door. The "TWO SOURCES OF SUPPLY" label is of limited use to a responder who must open a door into an energized 480 V compartment to operate either device. PB1 is an abort button — it commands the opposite of shutdown.

R-35 — Terminal-number collision: T20 and T21 are printed on two different circuits, one of them a Class A fire SLC return. *drafting (verifier-added).* Sheet 6's own numbering scheme (7 per cabinet) allocates cab3 = T15–T21, and sheet 7 accordingly labels TB10 "TB SLC CAB3 — TERMS 15/16 + 20/21" (`SLC_R3P/N` to FACP1 pins 10/11). But sheet 6 labels TB20 "TB LOOP CAB1 — TERMS 20/21" and the ARCH E says "TB LOOP CAB1 (T20-21)" (`LOOP1_P/N`). Two physically different pairs carry the same printed numbers — a wire-to-the-wrong-terminal risk on a life-safety circuit. The loop blocks should start at T22.

R-36 — The A3 root sheet carries no connectivity and the set has no inter-sheet signal cross-references. *drafting (verifier-added).* The seven hierarchical sheet symbols on sheet 1 are empty rectangles with no sheet pins; every cross-sheet signal is a bare global label with no "continued on sheet n / zone x" origin-destination reference at either end (`FACP_AC_L` on 3 and 7, `DC_COM` on 3/5/6, `PE` on 2/3/4/6). Tracing any signal across sheets requires a full-text search of the PDF. This is the largest practical obstacle to using the A3 set for construction.

R-37 — Lettering height: 94.7% of the A3 set is below the 2.5 mm minimum; the revision letter itself is 2.09 mm. *drafting (DQ-08).* Measured from PDF character objects on a genuine full-scale A3 (1190.5 × 841.9 pt): 16,340 of 17,245 characters below 2.5 mm; component values and net labels at 1.77 mm; **pin names at 0.97 mm and pin numbers at 1.11 mm** — the terminal and pin information a panel shop actually needs is the least readable content on the sheet. ARCH E: 56% below minimum. Against ASME Y14.2's 3.0 mm for "all other characters" the A3 non-compliance is ~97%. At a half-size check print these become ~0.5 mm. The ARCH E has ~23% of its sheet height empty and can absorb the increase directly; the A3 set needs re-layout or more sheets.

R-38 — No terminal-block schedule, cable schedule, wire-number scheme or layout; and one terminal is drawn twice. *drafting (DQ-10).* No schedules of any kind ship in `pkgJ/`. The ARCH E — which the WireSpec nominates as "the authority" — drops the SLC terminal numbers (TB8/TB9/TB10) that the A3 set carries. `TB5.3` and `SB1.1` are the same node on the same net (same for TB6/SB2, TB7/SB3), so the drawings show 12 CAN positions while the BOM buys 9 — correct only if the reader knows they are the same physical terminal, which no note states. All of this data is already in `golden_nets.json` and can be generated.

MINOR — 21

ID	FINDING	DISCIPLINES
R-39	JX1_IN pin designation stated **three ways** for one circuit: K/L (netlist and field-box pin names), 3/4 (ARCH E), 1/2 (fire sheet). The buildable artifacts are right; the notes are wrong twice. The vendor lead **decoded the letter scheme** (JXH1 A/B/C/D = 1/2/3/4 anchors A=1; I/J then = MCAN 9/10, corroborated by both CATL documents) so K/L = 11/12 and pins 1/2 are a different circuit — this converts the open architectural question into a narrow documentation request, caveated on the 12P-vs-16P housing dispute.	vendor (VI-01, VI-20)
R-40	PT1: no PE pin on the symbol so the case is on no net; `MTR_N = [M1.4, PT1.8]` only, so the metering secondary is floating (IEEE C57.13.3-2014 cl. 5); no secondary overcurrent protection at all; and the per-unit ratio is deferred in three documents — a 480:120 unit at 277 V gives **69.3 V**, below the eGauge's ~85 V minimum. **The EG4115 is rated for direct 480Y/277 connection, so deleting PT1 removes a $430 line, three fuses, four nets, the floating secondary and the ratio question at once.**	power (P-14, P-15, verifier-added)
R-41	No neutral bar and no DC-common block. `AUX_N` needs 7 panel-side landings, `SEC_N` needs 5 off a single T2 XN screw, `DC_COM` has 15 with no block at all. NEC 110.14(A). GB1 is the only bus in 57 BOM lines.	power (P-16)
R-42	Q2's device class is stated three inconsistent ways: BOM "UL489 breaker", sheet 4 "UL508A DISCONNECT", symbol `DISC_3P_40A`. **The board rejects the 440.22(B) violation claim** — that was computed per pole; the feeder-level computation permits 45 A and 25 A is compliant. Defect is the ambiguity only.	power (P-07)
R-43	CAN shield modelled as landing at the cabinet end (`CANn_SHD = [SBn.1, TBn.3, Xn.3]`) contradicting the wire spec's "floats at the cabinet", against a CATL requirement for a single grounding point — while CATL's Table 3-2 lists no shield contact on JX1_IN at all, and no shield conductor terminates in any MX34 plug in the netlist despite BOM item 36 buying 12 shield crimps.	controls (CTL-06), vendor (verifier-added)
R-44	Crimp contacts attributed to **Molex**; M34S75C4F2/F3 are **JAE** (doc JAHL-1754), and the cited "Molex MX34JAHL SOP" does not exist. Also: `catl/cablesop.*` and `catl/connectors.*` are **byte-identical** — the package holds 6 distinct vendor documents, not 7, and every "CATL cable SOP" citation resolves to the same JAE manual. Plug part numbers also conflict between CATL documents (360242-00813/814 vs Recodeal REVL16P100A, 12P vs 16P housing) and the BOM attributes JONHON parts to "CATL/Recodeal".	drafting (DQ-05), vendor (VI-07, VI-09, verifier-added)
R-45	Scope-of-supply inversion: CATL states "Connectors are provided by CATL. External cables of rack and CANbridge are provided by customers," and the BOM books the adapter cables at $0/CATL and carries no CANbridge line.	vendor (VI-10)
R-46	Wire colours specified three ways for CAN (yellow/green vs blue/green vs white-by-omission, with green assigned to both CAN_L and the shield), and "HARNESS COLORS PER CATL: … N BLACK" sits on the same sheet as "NEUTRAL GREY". No CATL document in the record contains any wire-colour specification. **The board does not seat this as an NEC 200.6 violation** — the governing legend says grey — only as a conflicting instruction.	vendor (VI-05), drafting (DQ-14)
R-47	Fire field boxes X12/X13/X14 carry **no CATL connector or pin designation** for 18 life-safety terminations, while X20-X22 two sheets earlier do it correctly ("JX1.K_LOOP+ … PER	fire (FLS-19), vendor (VI-08)

ID	FINDING	DISCIPLINES
	EW V1.2"). CATL's acceptance checklist is written entirely in pin letters, so it cannot be executed against this drawing.	
R-48	Multiple-source label names two of four energy sources and does not say **where** the second is isolated (the Kehua aux output breaker inside PCS Cabinet-375) — precisely the field-terminal case 409.110(3) targets — and there is no NFPA 70E stored-energy/LOTO source list covering BAT1/BAT2 (24 V, 192 Wh, no fuse or disconnect in the netlist) and UPS1. **The board rejects the "409.110 marking absent" and "live line terminals are a defect" halves** — the marking exists, and TB1 poles 1-3 sit at 277 V after Q1 opens by identical logic.	power (P-09)
R-49	SCCR: the standing gate is already disclosed on sheet 2, but two points are new — NEC 409.110(2)/(4) requires the marking **per incoming supply circuit** and this panel has two, and the aux-side available fault is only **912–1,519 A** (45.6 A / 3–5% Z) against a 480 V side governed by Q1's unmarked 5 kA default. One blanket number is the wrong label.	code (CODE-05)
R-50	Two non-identical `golden_nets.json` files ship as "the source of truth", differing on 16 nets because the file is keyed by **ordinal pin index** and the ETH symbol has a different pin order in the two generators. Three of four scripts load the outG copy; **nothing in the delivered suite compares the two formats.** The chair notes the design itself is fine — name-keyed canonicalisation shows the two formats are electrically identical. Also `pkgJ/multi-sheet-A3/` ships **without** a golden file, so the A3 set cannot be re-verified from the package.	drafting (DQ-17, verifier-added)
R-51	The Altium project enrols both complete SchDocs with `NetIdentifierScope=Flat`, so compiling presents 196 components with 98 duplicated designators and merges identically named labels. Neither SchDoc has Altium title-block parameters.	drafting (DQ-18)
R-52	DC_COM is not bonded to PE and there is no insulation monitoring — and the decision is undocumented. Not a code violation (UL 508A mandates neither), but a Rev J recommendation to carry the "confirm before bonding" note forward was **accepted and then not implemented** in Rev J's own master sheet. Gate-6.	controls (CTL-13)
R-53	F17 is a single fuse for all three CATL ETH modules while each MBMU gets its own — a fault in one module's wiring removes telemetry from all three cabinets. Two fuse holders, ~$22. Carson label scheme carried forward without re-derivation.	controls (CTL-12)
R-54	Kehua Modbus map declares an application scope (BCS1725K / NXPCS / BCS200K) that **does not include** the purchased BCS125K; all units default to 192.168.28.240; and no IP schedule exists anywhere. Minimum polling interval 100 ms is not stated as an EMS constraint.	vendor (VI-14), controls (CTL-16)
R-55	Duplicate fuse purchase: BOM lines 4/5/6 buy 17 fuses for 17 positions, then line 13 buys 12 more of the same F6-F17 range — **29 fuses for 17 holders, $110.40**. Holder count (17) confirms which line is the duplicate. This also falsifies `README.txt`'s "BOM↔drawing ref audit clean both directions". Two different holder part numbers are named for the same fuses.	drafting (DQ-12)
R-56	Package README is titled Rev H, names a BOM filename that is not shipped, quotes 145/145 when the count is 148, omits 9 shipped files, and `bomH.py` writes Rev J content to a file named `BOM-RevH.xlsx`.	drafting (DQ-15)

ID	FINDING	DISCIPLINES
R-57	Title blocks: "STR Medical" as responsible party, internal title "ARCH D MASTER" on a true ARCH E sheet, sheet size field "User", no drawing number anywhere (so NEC 409.110(6) has nothing to point at), sheet-1 title duplicated and clipped, and **no DRAWN BY / CHECKED BY / APPROVED BY fields in either format** — a PE has no originator or checker block to sign against.	code (CODE-17), drafting (DQ-11, verifier-added)
R-58	Live Simulator ships with "verified Rev G engineering model", "Rev E model loaded (139/139 nets verified)" and a **PSU3 NDR-75-24** block label — the exact undersized supply Rev J replaced. It is the artifact most likely to be shown to a non-engineer. `alt_nettest.py` likewise still prints "reproduces the verified Rev I design intent".	drafting (DQ-16, verifier-added)
R-59	Naming and misc.: `SUP1/2/3` (the releasing circuits) collide with `FACP_SUP` (the supervisory relay); no SLC device schedule or point count against the ARC-100's 100 points, and the CATL gas-detector option changes the cross-zone criterion the Handbook and Simulator narrate — while CATL's own documents contradict each other on whether gas replaces the heat or the smoke detector; the JXH1 EGC (`X6.5/X7.5/X8.5`) has no documented termination although JXH1 is a 4-position connector; Handbook p.10 states "Modbus TCP over RS-485", which is incoherent; the 480 V source grounding configuration (solidly grounded wye) is never required to be confirmed although the SPD, PT1 and any slash-rated device all depend on it.	fire (FLS-15, FLS-16), vendor (verifier-added), drafting (verifier-added), code (verifier-added)

OBSERVATION — 9

R-60 The five verification scripts prove drawing↔netlist fidelity and nothing else. None checks `golden_nets.json` for completeness against the BOM — which is exactly why two purchased loads (R-03) can have no circuit while every script reports PASS and the ARCH E banner claims "COMPLETE PANEL SCHEMATIC". Add a sixth script that reconciles BOM refdes against netlist refdes in both directions, plus the cross-format comparator of R-50. *(power, controls, drafting)* **R-61** SPD lead length: the 0.5 m note is physically unverifiable because no layout drawing exists to dimension it. **The board explicitly rejects the reviewer's inference that schematic column assignment implies backplate separation** — it does not. The physics cited is correct (~500 V per foot at 10 kA 8/20 µs against a 1,200–1,800 V VPR) and belongs in the layout package. *(power P-11)* **R-62** No documented load allocation on the 30 kVA Kehua aux transformer: this panel is 11.0 kVA at PF 0.85 (37% of 45.6 A FLA), but Kehua publishes no PCS self-consumption figure and the same transformer serves three module cooling systems, cabinet controls and the EPO circuit. "<< 30 kVA" accounts for one of two loads. *(power P-17)* **R-63** Tap-rule compliance for the F1-F5 primary taps (240.21(B)(1)) and the T2 secondary conductors (240.21(C)(2), since the (C)(1) two-wire exemption does not apply to a 3-wire secondary) both pass, but a plan checker has to derive it. Two one-line notes remove the question — conditional on F4/F5 being defined per R-05. *(power P-18)* **R-64** VRLA batteries: 49 °C is the FACP's ambient *limit*, not a declared sustained battery ambient, so "sub-one-year life" is a worst-case bound, not a projection. The real gap is that no battery design ambient is stated, the 8.0 Ah 25 °C/20 h nameplate is never derated, and no replacement interval is set. *(fire FLS-17)* **R-65** Kehua aux secondary is never documented as a wye with an accessible, load-rated neutral — the string "220" does not appear in `pcs_spec.txt` — yet 100% of the 9.45 kW aux load is single-phase L-N off `AUX_N` from X18 pin 4. Largely inside the standing Kehua submittal gate; add one explicit line requesting the secondary winding diagram and neutral terminal size. Also unreconciled: three Kehua output breakers against one five-wire feeder. *(power P-05a, vendor VI-11)* **R-66** Builds are not reproducible: 1,909 UUIDs regenerate per build, so no checksum ties a delivered PDF to a delivered source. UUID-normalised, the packaged and working `.kicad_sch` are byte-identical. *(drafting DQ-21)* **R-67** Unverifiable provenance hedges: "(PUBLIC SHEET 32A)" cites a document not in the record (no `32A` appears in any CATL extract), and "(PUBLIC REV)" understates figures that in fact come from the controlled spec R08306P05L31 which **is** in the package. The hedge traces to a Rev D note still printing on the Rev J sheet 1. *(vendor VI-17)* **R-68** Designators used in documentation that do not exist on drawings: `ETH1-3` (drawings say EM1-3) and `CT1-9` (the nine CTs have an empty Ref Des cell and appear only as X15/X16/X17 pins), so nine purchased items have no identifier for installation labelling or as-built markup. Outdoor pad anchorage, wind and flood exposure are unaddressed but are site EPC / structural EOR scope; the in-scope residue is one row in a responsibility matrix. *(drafting DQ-20, code CODE-18, CODE-12)*

---

3. FINDINGS THE BOARD DEMOTED, RESCOPED OR CLOSED

Recorded so the record shows what was tested and did not survive.

- **"Missing 120 Ω MCAN terminator" (CTL-01, raised CRITICAL) — headline refuted.** CATL's own checklist requires the J902 9-12 / 10-11 **wire jumpers** to read "on" and expects ~60 Ω at J902.21/22, proving the bus is terminated at both ends without a customer-added resistor. Adding resistors at J902 would break the vendor acceptance test. Rescoped to R-33 (procurement + length budget only).
- **"Enclosure cooling at ⅓ of required" (P-04/CODE-07) — magnitude not seated.** Three independent recomputations gave 426 W, 566 W, 931 W and 1,300 W depending on absorptivity, shading and whether vertical-surface incidence is corrected. **The board seats the absent calculation, not a proven shortfall.** Correct statement: specified 1,000–1,500 BTU/h is marginal to roughly 2× undersized depending on shading and absorptivity, further eroded because catalogue capacities are published at L35/L35 and derate at the 36.7 °C ASHRAE 0.4% design ambient — and no calculation exists to tell which. (Folded into R-03's action list as a required deliverable.)
- **"Q2 violates NEC 440.22(B)" (P-07, raised MAJOR) — demoted to MINOR (R-42).** The ceiling was computed per pole; 430.62(A)/440.22(B)(1) are written for the feeder, which permits 45 A.
- **"Four energy sources, label addresses two" (P-09, raised MAJOR) — demoted to MINOR (R-48).** The 409.110(3) marking exists; energised line terminals upstream of a main are universal and the same condition exists on the 480 V side.
- **"SPD lead 0.5 m unachievable" (P-11, raised MAJOR) — demoted to OBSERVATION (R-61).** Schematic column ≠ panel layout.
- **"BMS branch AWG14 unterminable" (VI-02, raised MAJOR) — demoted to MINOR wording.** The AWG10 adapter cable is already specified three lines earlier; the transition is at TB2-TB4 and the panel-side AWG14 never enters a JXH1 contact.
- **"CATL requires >1500 VAC aux harness" (VI-03, raised MAJOR) — demoted.** The withstand value is byte-identical to the HV row (signature of a partially edited table) and CATL's own diagram labels these conductors AWG10, a 600 V building-wire designation. Specifying 2 kV DLO for a 220 V, 13.6 A branch is not sound. Carry the checklist row into the commissioning package with a written disposition — along with the CAN and fire rows, which have the same issue.
- **"CB2 common trip is a single point of failure" (CTL-11) — closed.** The proposed remedy would create a violation: `CTRL_L1`/`CTRL_L2` share `SEC_N`, making this a multiwire branch circuit under NEC 210.4(B),

which requires simultaneous disconnection at the origin.

- **"NAC appliances unsupervised" (FLS-06, raised MAJOR) — demoted to MINOR.** Both appliances are inside the enclosure; the unsupervised segment is about a metre of in-panel wire behind a closed door. The 4-pin symbol fix is still correct.
- **"No notification at the protected hazards" (CODE-14) — refuted.** Each EnerOne+ cabinet has its own sound-and-light alarm on both first- and second-level alarm.
- **"NEC 705.13 Power Control System listing" (CODE-12) — not seated.** No export-limiting function is stated anywhere in the set, so the listing trigger is speculative. The responsibility-matrix ask survives as R-68.
- **"R1-R3 duplicate termination" (VI-19), "JX1_IN K/L vs 1/2 ICD conflict", "SCCR determination", "FSS option A vs B", "Kehua submittal", "Battalion VLAN sign-off", "PE stamp", "NFPA 72 final battery calc" — standing gating items, not re-reported as new.** Where new content attached to one of them it is seated separately (R-16, R-33, R-39, R-49, R-65).
- **"Enclosure A/C 225% overloads T2" (P-01, raised CRITICAL) — demoted to MAJOR (R-03).** The arithmetic is right but describes a circuit that does not exist. The reproducible defect today is the per-leg census, which the reviewer missed and which the board seated instead.

---

4. BOARD VERDICT

(1) INTERNAL CORRECTNESS — the drawings faithfully and consistently express the netlist. They do not consistently express the design.

The automated result is genuine. The chair independently regenerated the KiCad netlist and re-ran the tests: **148/148 matched, 0 failures, 98 components, 2 intentional spare-port no-connects.** Four leads reproduced the same, plus `alt_nettest` 148/148, `alt_nettest_simple` 148/148 and `audit_binary` 0 violations over 3,299 CFB frames. One lead went further and canonicalised both extracted netlists by pin **name** rather than ordinal index and confirmed the ARCH E master and the 8-sheet A3 set are **electrically identical** — 148 nets, zero membership differences.

What 148/148 proves: that the wire geometry drawn on the ARCH E sheet, on the 8-sheet A3 set, and in both Altium editions reproduces `golden_nets.json` exactly. That is a real and unusual achievement and it eliminates an entire class of transcription error.

What 148/148 does not prove, precisely: - **It does not test the netlist against the BOM.** No script reconciles purchased items to refdes. That is exactly why an air conditioner and a heater (R-03), three EOL devices (R-13), a PSN-1000 (R-12) and an LTE router (R-32) can be bought while every test reports PASS and the ARCH E banner reads "COMPLETE PANEL SCHEMATIC." - **It does not test the annotation layer.** Every note, value string, label, VLAN table, colour rule and revision statement is outside the tested surface. That is where R-06, R-07, R-19, R-20, R-35, R-39 and R-46 live — including a live instruction to create a second N-G bond and a terminal number printed on two different circuits. - **It does not test that the netlist is buildable.** Two mutually exclusive FSS architectures are both live and both counted in the certified 148 (R-10). A logical netlist with 15 landings on `DC_COM` and 5 on a single transformer terminal passes (R-41). - **It does not compare the formats to each other.** Two non-identical `golden_nets.json` files ship as the source of truth, differing on 16 nets, and no delivered script would catch a real divergence (R-50). - **It says nothing about sizing, coordination, listing, marking, or vendor conformance.** Every finding in section 2 passes all five tests.

Beyond the netlist, internal consistency across formats is **not** achieved: the A3 index sheet describes a Rev C machine under a Rev J title block, the sheet-2 aux-source statement contradicts sheet 4, both governing drawings specify two different breakers for one circuit, the ARCH E and the A3 disagree on TB11's terminal type and on the SLC terminal numbers, and the fire sheet, the ARCH E and the netlist name three different connector pins for one circuit.

(2) ENGINEERING ADEQUACY — the intended design is right in the signal and comms domain, materially incomplete in the power train, and not adequate in the fire-alarm domain.

What is genuinely right, and was attacked and held: conductor ampacity throughout, with correct 310.15(B)(1) / 110.14(C) / 240.4(D) treatment at a 49 °C internal ambient (8 AWG feeder passes with 13% margin; AWG10 aux branches with 50%; 14 AWG BMS branches correctly at 6 A). The #8 N-G bond meets Table 250.102(C)(1) and there is exactly one bond per separately derived system in the netlist. The 333 mV UL 2808 CT decision correctly eliminates the open-secondary hazard. Phase balance across the aux distribution. Rail A redundancy is real at the supply level (35% loaded with one supply failed) and UPS1's ~100 s claim is honest for the load it actually backs. CAN topology is a correct per-cabinet daisy-chain with no stubs, correct polarity end to end, and the MBMU/ETH pin mapping matches CATL's checklist on all three channels. The Rev J USLKG fix is correct where it was applied. The chiller breaker **value** (20 A) is correct even though its stated code basis is not. Rev J's PSU3 upsize is correct arithmetic. All 51 applicable CATL Cable Connection Checklist items pass against the netlist.

Where the intended design is not adequate:

Fire-alarm architecture is the weakest domain and it is the one where the consequences are highest. There is no hardwired path from the fire panel to the inverter (R-01); the fire panel is on a shared branch with no code identification (R-02); the Class A pathway is defeated for the damaged raceway (R-11); the three releasing circuits are drawn onto the wrong pin function at 114% of the panel's circuit rating with the required aux supply carried only as a BOM contingency (R-12); three purchased supervision devices are not in the design (R-13); the abort function is a single unsupervised pair for three hazards (R-14); there is no manual release and no maintenance disable (R-15); no detection at the control unit (R-27); and no listed off-premises transmission (R-28). Several of these were half-seen by the Rev J swarm and closed with a note. **A note is not a design.**

The power train has a structural hole and a specification hole. Two purchased thermal loads have no circuit and the 500 VA CPT has no per-leg census (R-03). Fourteen of seventeen fuses cannot be bought, the CPT primary protection ceiling (3 A per 450.3(B)) is nowhere stated, and the specified fuse class is fast-acting where time-delay is required (R-05). Six branch breakers on a 380Y/220 V system are procured with no voltage rating and no interrupting rating (R-04), and thirty of fifty-seven BOM lines have no orderable part number, including every overcurrent device (R-31). The aux feeder's protection method is stated two contradictory ways (R-18). The 480 V train remains sized for a transformer deleted a revision ago (R-07).

Redundancy is inverted and unmonitored. $645 of N+1 protects 26 W of EMS while every safety-relevant load hangs off one unbacked supply (R-08), and no supply-health signal reaches anything, so the first failure is silent (R-09). UDGOK gates 3 and 4 both fail on the rail that matters.

Quantification (gate 7) fails in four named places: no 24 VDC load schedule (R-22), no enclosure heat balance with solar gain (R-03), no fuse coordination study (R-23), no aux transformer load allocation (R-62).

Vendor conformance is good at the pin level and weak at the specification level. Every pin assignment checks out against CATL's own acceptance test. What does not: the rack-end terminator is not procured (R-33), the loop conductor gauge is self-contradictory (R-24), the Modbus map does not cover the purchased model (R-54), the crimp contacts name the wrong manufacturer and a non-existent SOP (R-44), and the scope-of-supply split is booked backwards (R-45).

One thing the board wants recorded on its own: two customer-facing documents make claims this design cannot support — "cellular backup takes over in seconds" (R-32) and "ring-wired detectors survive a cut cable" (R-11). Those must be corrected in the same revision as the design, not after.

(3) RELEASE READINESS — not ready for construction. Ordered gates, with owners.

Gate A — Close before any further drawing work (design decisions that change sheets).

#	ITEM	OWNER
A1	Design the hardwired fire→PCS safe-state interlock: FACP relay → interposing relay → Kehua remote-stop; re-polarise the three FACP inputs to N/C with supervision; split `FACP_COM`; move fire-path I/O to a backed rail. Add to netlist. **(R-01)**	Panel design EOR + fire alarm designer
A2	Add the dedicated fire alarm branch circuit off `SEC_L1` with red identification, lock-on and OCPD placard. **(R-02)**	Panel design EOR
A3	Select FSS Option A or B and delete the losing architecture from the live netlist. Everything in R-12, R-13, R-14, R-24 and the R-16 battery term depends on this. **(R-10, standing gate)**	Client + AHJ + CATL/Potter ICD
A4	Restructure the 24 V power tree so a single supply failure cannot remove BMS and fire I/O, and add supply-health points. **(R-08, R-09)**	Panel design EOR
A5	Add the enclosure A/C and heater as real circuits with a source and OCPD, and publish the T2 per-leg VA census. **(R-03, R-30)**	Panel design EOR
A6	Re-draw the SLC to CATL's native rack-to-rack chain with separated outgoing and return raceways, or re-declare Class B and delete the CA-6075. **(R-11)**	Fire alarm designer

Gate B — Vendor RFIs, issued now because Gate A and Gate C depend on the answers.

#	ITEM	OWNER
B1	Kehua, one letter: aux secondary configuration and neutral availability/rating; ampere, pole and interrupting rating of each of the three output breakers; N-G bond location; existence and contact form of a remote-stop/EPO input; PCS self-consumption; Modbus protocol revision covering BCS125K; number of northbound Ethernet endpoints; Modbus master-timeout behaviour. **(R-01, R-18, R-54, R-62, R-65)**	Procurement + panel design EOR
B2	CATL/Potter ICD, one letter: chiller nameplate MCA/MOCP and locked-rotor current; MBMU and ETH-module 24 VDC quiescent and peak; JX1_IN/JX2_OUT contact wire range and the full 12-position pinout (with the 12P-vs-16P housing question); the terminal-resistor connector part number and who supplies it; aerosol initiator minimum actuation voltage; which detector option is ordered and CATL's own heat-vs-smoke contradiction; JXH1 neutral separation; the cable-type checklist rows (aux, CAN, fire) disposition. **(R-19, R-22, R-24, R-33, R-39, R-59)**	Panel design EOR

Gate C — Produce the missing controlled documents. None of these exist today; none can be waived.

#	DELIVERABLE	OWNER
C1	Fuse and breaker schedule (refdes, class, amps, voltage, IR, code basis), including F4/F5 ≤ 3 A time-delay and the 450.3(B) derivation for T2. **(R-05)**	Panel design EOR
C2	Protective-device specification with full part numbers, voltage ratings and interrupting ratings for all 10 breakers, resolving the UL 489 vs UL 1077 question on Q1 and CB1. **(R-04, R-31)**	Panel design EOR + procurement
C3	Label and nameplate schedule: 409.110(1)–(7) per supply, 409.22(B) field marking, 110.16 arc-flash, 110.22(A) purpose identification, fire alarm circuit labels, ESS signage, LOTO source list. **(R-17, R-48)**	Panel design EOR
C4	Enclosure layout / elevation with the 760.136 fire-circuit segregation method, 110.26 working space, 312.6 bending space, SPD lead envelope, and door-device locations. **(R-25, R-34, R-61)**	Panel design EOR
C5	Enclosure heat balance: stated design ambient, effective area, solar gain by face, A/C derating at 36.7 °C, resulting BTU/h with margin — then re-select the A/C. **(R-03)**	Panel design EOR
C6	24 VDC load schedule + fuse coordination study against the real supply current-limit and TCC curves. **(R-22, R-23)**	Controls EOR
C7	I/O point list, alarm point schedule, Modbus register/setpoint schedule with watchdog interval and failsafe setpoint, and the port/VLAN/IP schedule as a signed sheet with a stored switch configuration. **(R-20, R-21)**	Controls EOR
C8	Terminal-block schedule, field cable schedule and wire-number scheme — generated from `golden_nets.json`, which already holds the data. Fix the T20/T21 collision in the same pass. **(R-35, R-38)**	Panel design EOR
C9	Single signed NFPA 72 secondary-supply worksheet, referenced by number from drawing, BOM and Handbook, resolving 3.3 vs 4.8 Ah and incorporating CATL's 2 h notification term. **(R-16)**	Fire alarm designer
C10	SCCR determination per incoming supply circuit with a UL 508A SB4 worksheet. **(R-49, standing gate)**	Panel shop (UL 508A) + panel design EOR
C11	Responsibility / code-compliance matrix (NEC 705/706, IEEE 1547, UL 9540/9540A, HMA, separation distances, commissioning and decommissioning plans, ESS signage, revenue metering split, pad anchorage), one row per requirement, one column per party. **(R-68)**	Panel design EOR + site EPC/EOR

Gate D — Drawing hygiene and change control, before re-issue.

D1 Rebuild A3 sheet 1: real revision-history table, current open items, banner driven from the same variable as the title block. D2 Delete every stale Rev C/I string — pad TX, TB12, "N-G bonds on BOTH transformers", "EXTERNAL PAD TX: N-G BOND + GEC AT TX", the 15 A C-curve sentence, "(PUBLIC SHEET 32A)", "(PUBLIC REV)", "Rev H design", "ARCH D MASTER" — and add a build-time assertion that every refdes token in note text exists in `golden_nets.json` and that any device rating named in a note matches the component value. **(R-07, R-19, R-67)** D3 Assign a drawing number series, carry it into every title block and the 409.110(6) field, and add DRAWN/CHECKED/APPROVED blocks. **(R-57)** D4 Raise lettering so pin names and numbers are ≥2.5 mm and values ≥3.0 mm. **(R-37)** D5 Add sheet pins to the root sheet and origin-destination cross-references to every cross-sheet signal. **(R-36)** D6 Publish exactly one name-keyed `golden_nets.json`, add the BOM↔netlist reconciler and the cross-format comparator, and seed UUIDs deterministically with a checksum manifest per issue. **(R-50, R-60, R-66)** D7 Correct the two customer-facing documents (LTE, ring-wired detectors) in the same revision as the design change. **(R-32, R-11)** D8 Delete BOM line 13 ($110.40 double-count), correct the JAE attribution, split the scope-of-supply lines, and rewrite the README. **(R-55, R-44, R-45, R-56)**

Gate E — Then, and only then. Independent re-verification of the corrected set (all six scripts including the two new ones), followed by review and seal by a Texas-licensed Professional Engineer, then issue for construction with the commissioning matrix attached (60 Ω at J902.21/22; ~230 V at JXH1 A/B and C/D; the CATL checklist rows mapped one-to-one onto our drawing; the R-33 terminator continuity check).

---

5. STANDING STATEMENT ON THE STATUS OF THIS BOARD

This review board is a set of AI review agents applying PE-level rigor. It is not a licensed professional engineer, and nothing in this document is a professional engineering opinion, a certification, or an approval. No finding here — including the ones marked CONFIRMED and independently re-derived from the source files — substitutes for review and sealing by a Texas-licensed Professional Engineer. Every code citation, calculation and vendor interpretation in this register must be independently verified by that engineer before it is relied upon. The board does not certify; a named human signs.